

Keerthi Raj Vasireddy Yuvaraj

Machine Learning Engineer

New York city, NY Open for relocation to Seattle, WA Dallas, TX, Jersey City, NJ Chicago, IL

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SUMMARY

Innovative Machine Learning Engineer with over 5 years of experience in developing and deploying AI-driven solutions, specializing in Generative AI and Natural Language Processing. Proven track record in building scalable ML pipelines, fine-tuning large language models, and integrating AI solutions into customer-facing applications. Adept at collaborating cross-functionally to drive product innovation and enhance user experience.

PROFESSIONAL EXPERIENCE

Machine Learning Engineer, Shopify

January 2024 - Present | New York, NY

- Led the development of *Shopify Magic*, a Generative AI platform that automates the creation of product descriptions, FAQs, marketing emails, and store metadata using fine-tuned LLMs tailored for e-commerce applications.
- Enhanced NLP pipelines utilizing Python, spaCy, NLTK, and PyPDF2 to extract and classify structured data from unstructured product inputs, improving metadata tagging accuracy by 30%.
- Streamlined end-to-end ML pipelines deployed on GCP Vertex AI and integrated with Streamlit, facilitating real-time AI-driven content suggestions within Shopify Admin and mobile interfaces, reducing merchant content entry time by 70%.
- Optimized product taxonomy classification models to auto-suggest categories and attributes (e.g., size, color, style) with over 92% accuracy, decreasing manual tagging errors by 60%.
- Improved LLM API integrations through prompt engineering and real-time user feedback loops, enhancing text relevance and tone, leading to a 75% satisfaction rate among merchants using AI-generated content.
- Collaborated with cross-functional teams to align AI-generated content with SEO strategies and brand guidelines, employing A/B testing frameworks to boost engagement on product and marketing pages.

Machine Learning Engineer, HCL Technologies

Jan 2020 – Sep 2022 | India

- Engineered machine learning models to optimize customer segmentation, improving marketing strategies by 20%. Collaborated with teams to enhance model accuracy.
- Automated machine learning pipelines, reducing manual work by 30% and speeding up model deployment. Streamlined training processes for efficiency.
- Developed deep learning models for unstructured data, improving processing speed by 25%. Integrated models into existing workflows for automation.
- Implemented real-time predictive analytics, enabling 30% faster decision-making. Collaborated with business teams to ensure effective integration.
- Optimized model performance through hyperparameter tuning and feature engineering, increasing prediction accuracy by 15%. Reduced processing times by 20%.

CORPORATE PROJECTS

Visual Question Answering System.

[Python, TensorFlow, Keras, GCP, Streamlit]

- Boosted model accuracy by 10% through hyperparameter optimization, analyzing 400,000+ imageQA dataset
- Automated model checkpointing on GCP, reducing GPU waste by 40% (\$200+/month cost savings)
- Deployed a Streamlit interface serving daily users, with continuous uptime through GCP load balancing

Gen AI Resume Screener.

[GCP, Vertex AI, Gen AI, MLOps, Spacy, NLTK]

- Built end-to-end pipeline using Python (PyPDF2, spaCy, NLTK) to extract and analyze 20+ data fields from Resume's with 95% parsing accuracy
- Processed 1,000+ Resume's using custom LLM prompts, achieving 65% screening accuracy and reducing manual HR review time by 100+ hours
- Designed an interactive Streamlit-based UI for Resume feedback, achieving 75% user satisfaction based on survey results

TECHNICAL SKILLS

Programming: Python, Java, C/C++, MATLAB, SQL

Machine Learning: TensorFlow, scikit-learn, Hugging Face, SpaCy

Deep Learning Models: BERT, LSTM, CNN, YOLO, ResNet MLOps: GCP Vertex AI, AWS SageMaker, Model Retraining Pipelines

Data Engineering: GCP Data Fusion, Apache Airflow, ETL

Pipelines Cloud Platforms: Google Cloud Platform (GCP), Amazon Web Services (AWS)

Cloud Services: AWS S3, GCP Cloud Storage

Monitoring & Logging: Python Logging Scripts, Real-time Monitoring Systems

Data Visualization: Power BI, Matplotlib

Feature Engineering: OpenCV Model Development

Environment: Jupyter Notebook

Collaboration Tools: Confluence, Jira

Version Control: GitHub

Deployment: CI/CD Pipelines (basic with GitHub Actions)

EDUCATION

University of Texas, TX, USA

Master of Science in Computer Engineering

Presidency University, Bangalore, India

Bachelor of Technology in Computer & Data Science